

The after-tax accumulations of the various investments are:

Year	Bond Fund	Non-Deductible IRA	Equity Fund	Non-Dividend Paying Stock	Tax Exempt
0	1000.00	1000.00	1000.00	1000.00	1000.00
10	2013.22	2271.93	2398.83	2604.66	3105.85
20	4053.07	6222.36	5754.36	7588.48	9646.29
30	8159.74	18491.79	13803.71	23067.46	29959.92

The non-dividend paying stock dominates the bond fund, the non-deductible IRA, and the equity fund. It is second only to the tax exempt strategy. It works so well because of the lower capital gains rate. From this perspective Warren Buffet is right not to pay dividends. But so many companies do!⁴ Why? The standard explanations are dividend clientele and signaling hypotheses, since paying dividends by itself loses money for shareholders. There may also be benefits to paying out cash rather than managers wasting it on white elephant projects in the firm.⁵

It is interesting that the non-deductible IRA and the equity fund switch positions between years 20 and 30. For short horizons, the equity fund dominates. For example, at the 10-year horizon, the equity fund cumulates to \$2,399 versus \$2,272 in the non-deductible IRA. So if you're saving for a short time period, forget the Roth IRA; just put your money directly into a mutual fund. But at longer horizons, the non-deductible IRA comes out ahead. At 30 years, the non-deductible IRA cumulated amount is \$18,492, which trounces the \$13,804 in the equity fund. (If you're twenty-five years old and reading this book, sock the maximum permissible into your non-deductible IRA every year.) The intuition for this is that the non-deductible IRA has earnings that compound at the higher pre-tax rate of 12% (see equation (12.2)), whereas the mutual fund compounds at the after-tax rate of 9.1% (see equation (12.3)). This is offset in the short run by the higher income tax rate for non-deductible IRAs and the lower capital gains rate for mutual fund investments. But over time, the "interest on the interest" dominates in the non-deductible IRA. Such is the beauty of compounding.

Clearly, taxes matter. This simple example shows that:

1. The most important lesson is to save in a way that is sheltered from taxes. Everything else is secondary.

⁴Fewer companies now are paying dividends but those that do pay out big ones. In fact, while the number of firms paying dividends has dropped over the past few decades, the sum of dividends paid by all companies has increased in both nominal and real terms. See Fama and French (2001) and DeAngelo, DeAngelo, and Skinner (2004).

⁵See Allen and Michaely (1995) for a literature summary, which is a little dated but still relevant.